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Outstanding performance of variable-speed pumped storage: Evaluation of operational characteristics of pumped storage units in high-proportion renewable energy systems

Yunpeng Zhang [a,b], Gongcheng Liu [c,*], Zhang Liu [a,b], Meng Zhang [a,b], Jianling Li [a,b], Ziwen Zhao [a,b], Ali Najah Ahmed [d], Diyi Chen [a,b,**]

[a] Key Laboratory of Agricultural Soil and Water Engineering in Arid and Semiarid Areas, Ministry of Education, Northwest A&F University, Shaanxi, Yangling, 712100, PR China
[b] Institute of Water Resources and Hydropower Research, Northwest A&F University, Shaanxi, Yangling, 712100, PR China
[c] College of Water Conservancy and Civil Engineering, Shandong Agricultural University, Shandong, Taian, 271018, PR China
[d] Department of Engineering, School of Engineering and Technology, Sunway University, No:5, Jalan Universiti, Bandar Sunway, 47500, Selangor Darul Ehsan, Malaysia

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ABSTRACT

The integration of a high share of renewable energy into the grid has transformed the flexibility requirements of modern power systems. The deployment of variable-speed units (VSU) to enhance the flexibility supply capacity of power stations has emerged as a key development direction for pumped storage (PS) systems. However, existing studies still lack sufficient evaluation of the operational characteristics of VSU under flexible operation modes. Therefore, this study develops a multi-energy complementary unit commitment model that reflects the flexible operational characteristics of PS units. Subsequently, an integrated benefit evaluation system is established using the entropy weight-TOPSIS method. Finally, a clean energy base in Qinghai Province, China, is analyzed as a case study to compare operational characteristics and quantitatively assess the flexibility performance of fixed-speed units (FSU) and VSU. The results indicate that for every 10 % increase in wind power penetration, the system's power supply reliability decreases by 1 %, while the total number of adjustments and cumulative power regulation of PS units increase by 5 %. Furthermore, VSU can effectively mitigate the adverse effects of higher wind power penetration on power transmission quality. VSU reduces the number of start-up and shutdown operations by more than 15 %, although it exhibits a higher cumulative number of power regulations. The average system flexibility indicator score for VSU increases by 21.26 points, while the average scores for operational characteristics of regulation units and system economic indicators improve by 10.64 and 9.04 points, respectively. This study provides an important reference for evaluating the techno-economic benefits of VSU under flexible operation modes.

1. Introduction

1.1. Background

China's determination to achieve carbon neutrality has accelerated the green transformation of its clean energy system. However, the randomness and intermittency associated with integrating high proportions of renewable energy pose significant challenges to power system operational security [1,2]. Pumped storage (PS) units, characterized by excellent start-stop performance and strong ramping capabilities, are recognized as ideal resources for reshaping wind and solar outputs and meeting energy system regulation needs [3,4]. Traditional fixed-speed units (FSU) have inherent limitations due to fixed-speed regulation, restricting their ability to fully adapt to substantial load fluctuations. Conversely, variable-speed units (VSU) enable flexible power output control through adjustable rotational speeds, unlocking deeper regulation potential and significantly enhancing flexibility [5,6]. Nevertheless, the deployment of VSU in China remains at an early stage, and fully realizing their flexibility advantages is currently a major research focus.

Meanwhile, to meet the accommodation demands of large-scale

* Corresponding author.
** Correspondence to: D. Chen, Institute of Water Resources and Hydropower Research, Northwest A&F University, Shaanxi, Yangling, 712100, PR China.
E-mail addresses: gongchengliu@sdau.edu.cn (G. Liu), diyichen@nwsuaf.edu.cn (D. Chen).

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wind and solar integration, the construction of PS plants in China is shifting strategically from eastern load-center user-side locations to renewable-rich western regions. Their primary role is transitioning from traditional peak shaving and valley filling to mitigating renewable energy fluctuations, requiring greater flexibility from PS units. Therefore, evaluating the operational characteristics of PS units under flexible modes has become crucial. This leads to the following key research questions:

1. How can the differences in operational characteristics between FSU and VSU be compared under flexible operation modes?
2. How can the advantages and associated costs of the enhanced flexibility provided by VSU be quantitatively evaluated?

1.2. Literature review

Flexibility provision capability refers to the ability of PS units to rapidly and effectively respond to dynamic changes and uncertainties within power systems. Unit flexibility is a critical indicator of operational characteristics, and adequate flexibility is essential for successfully integrating intermittent renewable energy into the grid [7]. This section reviews the flexibility characteristics of PS units, discusses the advantages of VSU along with existing research, and identifies current research gaps.

PS units significantly enhance power system flexibility. He et al. proposed a transformation approach for cascade hydropower-VRE systems, demonstrating that converting conventional hydropower stations to PS effectively improves system flexibility [8]. Luo et al. revealed that integrating PS units into cascade stations significantly reduces peak-valley differences in grid residual load, enhancing grid load-following capabilities [9]. Zhang et al. developed a soft-link coupling model providing an optimized framework for flexible PS unit regulation [10]. Previous studies have validated the technical and economic feasibility of pumped storage (PS) in multi-energy complementary systems from various perspectives [11]. Hozouri et al. emphasized the essential role of PS units in reducing wind curtailment and balancing supply-demand via a wind-storage coordination model [12]. Yuan et al. optimized short-term operation strategies for wind-storage complementary systems, smoothing the peak-valley difference and increasing wind penetration [13]. Basu et al. demonstrated PS advantages in managing wind and solar uncertainties within hydro-wind-solar-storage hybrid systems [14]. Xia et al. proposed multi-timescale optimal scheduling strategies based on wind-solar-thermal-storage complementary operations, confirming the significant contribution of PS to improving overall system regulation [15]. Wang et al. assessed the influence of intermittent energy absorption rates on PS regulation intensity, showing that every 2 % increase in absorption rate raises regulation intensity by 5 % [16]. Li et al. demonstrated the positive impact of PS on reducing wind curtailment and carbon emissions through a hybrid power system scheduling model [17].

However, traditional FSU still suffer from limitations in power response speed and flexible regulation capability, making them less effective in managing the rapid dynamics associated with high-penetration renewable energy integration. In response, researchers have increasingly focused on the development of VSU. Zhao et al. assessed the impact of five flexible PS technologies on energy systems [18]. Yang et al. demonstrated that VSU significantly outperform FSU in power response speed and suppression of wind power fluctuations [19]. Li et al. proposed a coordinated capacity optimization method for wind-storage hybrid systems, verifying the complementary operational advantages of VSU and wind power [20]. Yao et al. developed a capacity optimization model based on stochastic programming, confirming VSU notable advantage in reducing wind curtailment [21]. Xu et al. further validated the flexibility benefits of VSU in frequency control across different grid scenarios [22]. Xu et al. also proposed an optimal operation strategy for VSU, confirming its substantial advantages in enhancing grid flexibility and operational efficiency [23].

Wu et al. highlighted that coordinated operation between VSU and battery storage can significantly improve system flexibility and reduce battery storage capacity requirements [24]. Although VSU requires a higher initial investment, it significantly reduces long-term system operating costs [25]. Li et al. examined VSU performance in power markets, confirming its superiority in both economic efficiency and flexibility [26]. Chazarra et al. further demonstrated that incorporating secondary regulation services can substantially shorten VSU investment payback period [27]. Teng et al. analyzed the impact of upgrading FSU to VSU on the total cost of future European power systems, indicating a potential 10–20 % increase in long-term benefits [28]. However, the above studies still present the following limitations: (1) The analysis of the operational characteristics of PS units remains insufficient. Most existing literature focuses on the role of PS units in addressing issues such as wind curtailment and carbon reduction in multi-energy complementary systems, while in-depth research on how variations in wind and solar capacity ratios specifically affect their operational characteristics is lacking. (2) A comprehensive evaluation framework has yet to be established. Current studies primarily assess the technical and economic advantages of VSU from a single perspective or based on a limited number of indicators. There remains a lack of systematic and quantitative evaluation of the flexibility advantages and overall benefits of VSU under flexible operation modes. These issues warrant further in-depth research and exploration.

1.3. Contributions

To address the above research gaps and practical engineering needs, this study develops a multi-energy complementary system unit combination model and examines the impact of FSU and VSU on power system flexibility, using a clean energy base in Qinghai Province, China, as a case study. The main contributions are as follows:

1. Considering the differences in regulation characteristics between VSU and FSU, a multi-energy complementary system unit combination model is established to reflect the flexible operation characteristics of PS units.
2. A systematic comparative analysis of the flexibility performance and operating characteristics of FSU and VSU in clean energy bases with varying wind-solar capacity ratios.
3. A comprehensive evaluation framework based on the entropy weight-TOPSIS method is proposed to quantitatively assess the flexibility advantages and overall benefits of VSU by integrating system flexibility, economic performance, and operational characteristics of regulation units.

The paper is structured as follows: Section 2 presents the modeling methodology for the multi-energy complementary system unit combination and develops the evaluation indicator framework. Section 3 examines a case study of a clean energy base in Qinghai Province, China, analyzing how varying wind-solar capacity ratios influence power system flexibility. Section 4 provides conclusions, followed by discussions in Section 5.

2. Method

The overall modeling framework of this study is shown in Fig. 1. Section 2.1 outlines the modeling methodology for the multi-energy complementary unit combination model, aiming to minimize operating costs. Section 2.2 defines the operational and cost parameters for FSU and VSU. Sections 2.3 and 2.4 apply the entropy weight-TOPSIS method to evaluate the differences in operational characteristics and comprehensive benefits of FSU and VSU within the energy base, based on three categories of indicators: system economics, system flexibility, and regulation unit operating characteristics.

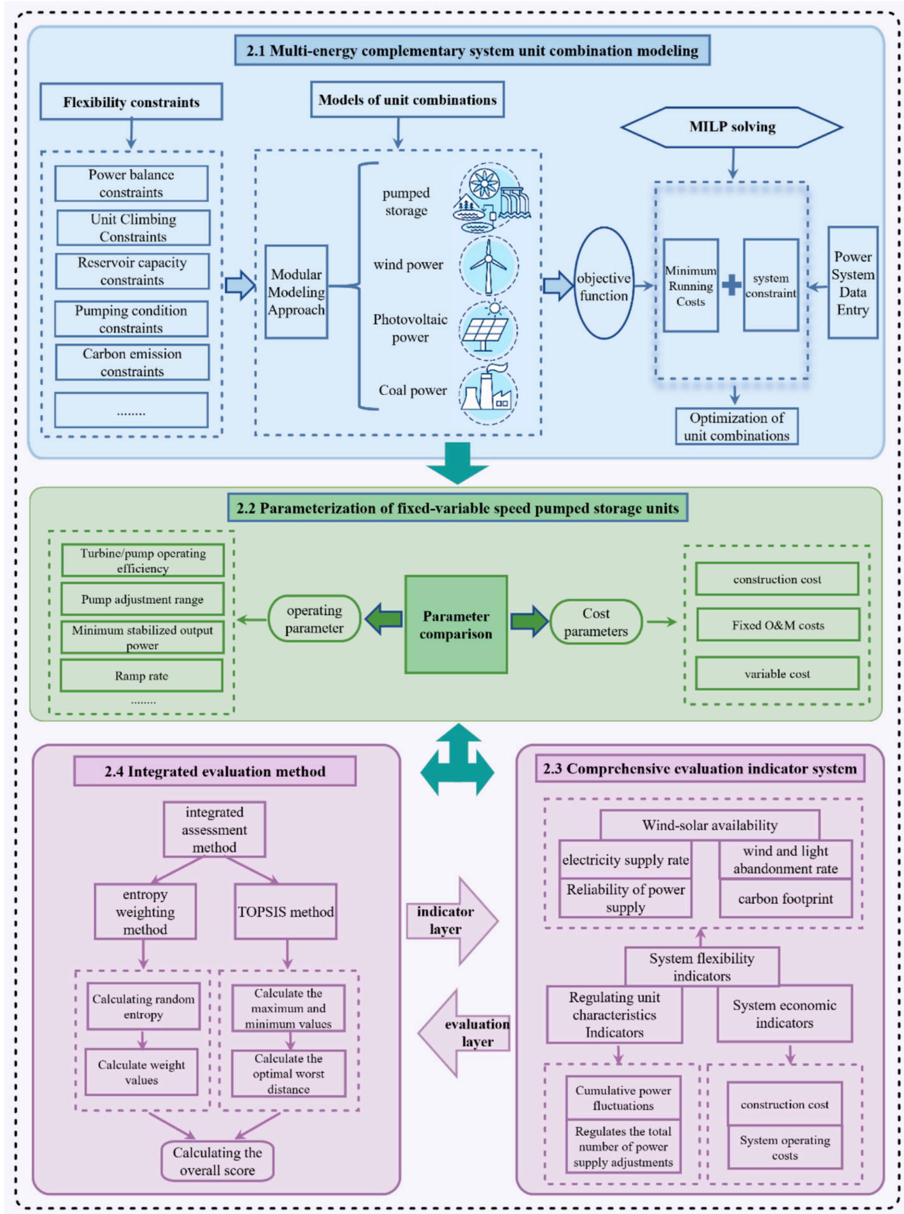


Fig. 1. A flow chart for the model development.

2.1. *Multi-energy complementary system unit combination model*

This section formulates a unit combination model to minimize system operating costs while reflecting the flexible operational characteristics of PS units. By configuring regulation parameters for different unit types, the model quantitatively evaluates the flexibility benefits of FSU and VSU and analyzes their operational differences. Mixed-integer linear programming (MILP) is employed to address the discrete nature of the unit combination problem in power systems.

2.1.1. *Objective function*

The multi-energy complementary system optimization and scheduling model aims to minimize the total operating cost over the scheduling period. The operating cost mainly includes the system fixed costs $C_{u,i}^{F}$, variable costs $C_{u,i}^{V}$, ramping costs $C_{u,i}^{RU}$ and $C_{u,i}^{RD}$, start-up and shut-down costs $C_{u,i}^{ST}$and $C_{u,i}^{SD}$, load shedding costs C_{i}^{LL}, transmission costs $C_{i,l}^{T}$, and curtailment costs C_{i}^{A} [29]. By summing these components, the objective function can be expressed as follows:

$$COS = \min\sum_{u,i} C_{u,i}^{F} + C_{u,i}^{V} + C_{u,i}^{ST} + C_{u,i}^{SD} + C_{u,i}^{RU} + C_{u,i}^{RD} + C_{i,l}^{T} + C_{i}^{LL} + C_{i}^{A} \quad (1)$$

where i and u are the time period and unit, respectively, and l is the transmission line.

2.1.1.1. *Fixed costs.*

The fixed cost mainly refers to the fixed investment and associated expenses incurred during system operation and maintenance, and is expressed as:

$$C_{u,i}^{F} = c_{u}^{F} \cdot x_{u,i} \quad (2)$$

where c_{u}^{F} denotes the fixed cost incurred per unit time when the unit is connected to the grid, and $x_{u,i}$ is a binary variable representing the on-grid status of the unit, taking the value 1 when the unit is in operation and 0 when it is offline.

2.1.1.2. *Variable costs.*

The variable cost refers to the cost directly associated with electricity generation and is given by:

$$C_{u,i}^{V} = c_u^{V} \cdot G_{u,i} \tag{3}$$

where c_u^V denotes the variable cost per unit of electricity generated, and $G_{u,i}$ represents the power output of unit u in time period i.

2.1.1.3. Start-up and shut-down costs. The startup and shutdown cost refers to the fuel, water, and other related expenses incurred during the startup and shutdown processes of generating units. In this study, two binary state variables, $y_{u,i}$ and $z_{u,i}$, are introduced to represent the startup and shutdown operations of the units, where $y_{u,i} = 1$ and $z_{u,i} = 1$ denote, respectively, the startup and shutdown actions of unit u in time period i:

$$\begin{cases} C_{u,i}^{ST} = c_u^{ST} \cdot y_{u,i} \\ C_{u,i}^{SD} = c_u^{SD} \cdot z_{u,i} \end{cases} \tag{4}$$

where c_u^{ST} and c_u^{SD} denote the unit startup cost and shutdown cost of the generating unit, respectively.

2.1.1.4. Ramping costs. The ramping cost is divided into upward and downward ramping costs, which are calculated as follows:

$$\begin{cases} C_{u,i}^{RU} \geq c_u^{RU} \cdot \left(P_{u,i} - P_{u,i-1}\right) \cdot x_{u,i} \\ C_{u,i}^{RD} \geq c_u^{RD} \cdot \left(P_{u,i-1} - P_{u,i}\right) \cdot x_{u,i} \end{cases} \tag{5}$$

where c_u^{RU} and c_u^{RD} denote the upward and downward ramping costs per 1 MWh of power change for the operating unit, and $P_{u,i}$ and $P_{u,i-1}$ represent the power outputs of unit u in time periods i and i − 1, respectively.

2.1.1.5. Transmission costs. The transmission cost represents the cost incurred by the system during power delivery and is formulated as:

$$C_{i,l}^{T} = c_l^{T} \cdot E_{l,i}^{T} \tag{6}$$

where c_l^T denotes the cost per unit of electricity transmitted on export line l, and $E_{l,i}^T$ represents the transmission power of line l in time period i.

2.1.1.6. Load shedding costs. The load-shedding cost refers to the cost incurred when the system fails to meet the transmission demand due to capacity shortages or other reasons, and is expressed as:

$$C_i^{LL} = c^{LL} \cdot E_i^{LL} \tag{7}$$

where c^{LL} denotes the cost per unit of unserved energy, and E_i^{LL} represents the amount of load shedding in time period i.

2.1.1.7. Curtailment costs. The curtailment cost represents the penalty cost incurred when the system output exceeds the transmission demand and power must be curtailed, and is given by:

$$C_i^{A} = c^{A} \cdot E_i^{A} \tag{8}$$

where c^A denotes the penalty cost per unit of curtailed energy, and E_i^A represents the amount of energy curtailment in time period i.

2.1.2. Constraints

The unit combination model imposes equality and inequality constraints on variables and parameters to simulate system operations. This section introduces the various constraints applied.

2.1.2.1. Power balance constraints. Power balance constraints ensure that, in any scheduling period, the sum of the system's power transmission demand, the pumping power consumption of PS units, and the curtailed energy equals the total generation of all units within the system plus the load shedding. This relationship can be expressed as follows:

$$\sum_u E_{u,i} + E_i^{LL} = \sum_l E_{l,i}^{T} + E_i^{A} + \sum_u E_{u,i}^{C} \tag{9}$$

where $E_{u,i}^C$ is the electricity consumption of PS units operating in pumping mode during time period i.

2.1.2.2. Unit output constraints. At any time period, under grid-connected operating conditions, the output of a unit must be greater than the specified minimum output limit and less than the allowable maximum output limit. This constraint can be expressed as follows:

$$P_u^{\min} \cdot x_{u,i} \leq P_{u,i} \leq P_u^{\max} \cdot x_{u,i} \tag{10}$$

where $P_u^{\min}$ and $P_u^{\max}$ are the minimum and maximum output limits of unit u, respectively, and the grid-connected operating status of the unit is indicated by a binary variable.

2.1.2.3. Ramping constraints. The stronger the ramping capability of a unit, the better the operational stability of the entire power transmission system. However, due to the physical limitations of regulation units, the change in output within a unit time must not exceed the unit's ramping capability, as expressed by:

$$\begin{cases} P_{u,i-1} - P_{u,i} \leq q_{u,i}^{-} \\ P_{u,i} - P_{u,i-1} \leq q_{u,i}^{+} \end{cases} \tag{11}$$

where $q_{u,i}^-$ and $q_{u,i}^+$ are the downward and upward ramping rates of the unit, respectively.

2.1.2.4. Reservoir volume constraints. The energy storage capability of PS plants is primarily constrained by the volume of the upper reservoir. At any time period, the actual reservoir volume must be between the maximum and minimum operational volumes for power generation. This constraint can be expressed as:

$$K_p^{\min} \leq K_{p,i} \leq K_p^{\max} \tag{12}$$

where $K_{p,i}$ is the actual reservoir volume of PS plants p during time period i, and $K_p^{\min}$ and $K_p^{\max}$ are the minimum and maximum operational reservoir volumes of PS plants p, respectively.

The power generation and pumping operations of PS plants must satisfy the reservoir power balance constraints. Considering that there is no natural inflow to the upper reservoir in this study, the reservoir volume at time period i is determined by the sum of the reservoir volume at time period i-1 and the pumped volume, minus the generated volume and the spilled volume at time period i. This relationship can be expressed as follows:

$$K_{p,i} = K_{p,i-1} + E_i^{C}\eta_c - E_i^{G}/\eta_g - q_i^{A} \tag{13}$$

where E_i^C is the pumped volume of PS plants at time period i, η_c is the pumping efficiency, E_i^G is the generated volume at time period i, η_g is the generation efficiency, and q_i^A is the amount of water discarded by the PS plants in time period i.

2.1.2.5. Reservoir water level constraints. The reservoir water level of PS plants at any time period must be between the minimum water level $L_p^{\min}$ and the maximum water level $L_p^{\max}$. This constraint can be expressed as:

$$L_p^{\min} \leq L_{p,i} \leq L_p^{\max} \tag{14}$$

where $L_{p,i}$ is the reservoir water level of PS plants p at time period i.

2.1.2.6. Pumping operation constraints. The main difference between FSU and VSU under the constraint conditions is reflected in the pumping operation constraints, while other differences are mainly captured through parameter settings. FSU cannot adjust input power during pumping operation, where as VSU is capable of flexible adjustment within a certain range. The pumping operation constraints for FSU and VSU can be expressed by the following two equations, respectively:

$$P_{u,i}^{C} = x_{u,i} P_{u}^{C,r} \eta_c \tag{15}$$

$$x_{u,i} P_{u}^{C,\min} \eta_c \le P_{u,i}^{C} \le x_{u,i} P_{u}^{C,r} \eta_c \tag{16}$$

where $P_{u,i}^{C}$ is the pumping power at time period i, $P_{u}^{C,r}$ is the rated pumping power of the unit, $x_{u,i}$ is a binary variable indicating whether the unit is in pumping operation or not, and $P_{u}^{C,\min}$ is the minimum pumping power of VSU.

2.1.2.7. Simultaneous generation and pumping operation constraints. In this study, the pumping and generation operations of FSU and VSU cannot be carried out simultaneously [30], and the constraint is expressed as follows:

$$\sum_{u} P_{u,i}^{G} \cdot \sum_{u} P_{u,i}^{C} = 0 \tag{17}$$

where $P_{u,i}^{G}$ denotes the generation power of PS unit u at time period i.

2.1.2.8. Coal power output constraints. During operation, the output power of coal power units must comply with their respective parameter limits and remain within a stable operating range. When a unit is offline, its output power is zero. The output constraints for coal power units can be expressed as follows:

$$T_{coal}^{\min} P_{u}^{coal} x_{u,i}^{coal,r} \le P_{u,i}^{coal} \le T_{coal}^{\max} P_{u}^{coal,r} x_{u,i}^{coal} \tag{18}$$

where $P_{u,i}^{coal}$ is the output power of the coal power unit at time period i, $T_{coal}^{\min}$ is the minimum output percentage coefficient of coal power, $T_{coal}^{\max}$ is the maximum output percentage coefficient of coal power, and $P_{u}^{coal,r}$ is the rated output power of large or small coal power units.

2.1.2.9. Minimum start-up and shutdown time constraints. After the coal power unit is started, it must operate for at least the specified minimum running time before it can be shut down. Similarly, once shut down, it must satisfy the minimum shutdown time requirement before it can be restarted. These time constraints are set to ensure the stability of unit operation and to prevent equipment damage caused by frequent start-ups and shutdowns. This constraint can be expressed as follows:

$$\begin{cases} \sum_{\tau=i-\gamma_u^{ST}-1}^{i} x_{\mathrm{u},\tau}^{ST} \le \mathrm{x}_{u,i} \\ \sum_{\tau=i-\gamma_u^{SD}+1}^{i} x_{\mathrm{u},\tau}^{SD} \le 1 - x_{u,\mathrm{i}} \end{cases} \tag{19}$$

where τ is the time period index, $\mathrm{x}_{\mathrm{u},i}$ is the grid-connected operating status of the unit, $\mathrm{x}_{\mathrm{u},\tau}^{ST}$ and $\mathrm{x}_{\mathrm{u},\tau}^{SD}$ are the start-up and shutdown operations of unit u at time period τ, respectively, and γ_{u}^{ST} and γ_{u}^{SD} are the minimum start-up and shutdown durations of unit u, respectively.

2.1.2.10. Carbon emission constraints. To comply with carbon emission limits under the dual carbon goals, the system's total carbon emissions must be lower than the specified threshold. This constraint can be expressed as:

$$CES = \sum_{u} P_{u,i}^{coal} k_{u}^{coal} \le F^{\max} \tag{20}$$

where $P_{u,i}^{coal}$ is the output power of the coal power unit at time period i, $F^{\max}$ is the carbon emission limit of the energy base, and k_{u}^{coal} is the carbon emission coefficient of the coal power unit.

2.2. Parameter settings

2.2.1. Operating parameters of PS units

The operating flexibility of generating units is determined by the combined influence of multiple parameters. Key factors affecting a unit's flexibility provision capability include maximum generation efficiency, minimum stable output, ramping rate, start-up and shutdown times. This section assigns corresponding parameter values according to the regulation characteristics of each unit type. Table 1 presents the relevant operating parameters for both FSU and VSU [18,31–34].

2.2.2. Cost parameters

Cost parameters mainly include the startup and shutdown costs, ramping costs, construction costs, fixed operation and maintenance (O&M) costs, and variable cost parameters for various generation and flexibility technologies, as shown in Table 2. The cost data for different generation technologies are sourced from the annual development report of china's electric power industry 2021 [35], while the costs for storage technologies are sourced from the development roadmap for large-scale energy storage technologies published by the Global Energy Interconnection Development and Cooperation Organization [36]. Compared with FSU, VSU incurs an additional 7 % in construction costs [34].

2.3. Evaluation indicators

To comprehensively evaluate the differences in operational characteristics between FSU and VSU within the clean energy base, the evaluation indicators are categorized into system economic indicators, system flexibility indicators, and regulation unit operational characteristic indicators. This section provides a detailed description of the definitions of each indicator and the corresponding calculation formulas.

2.3.1. System economic indicators

Construction cost (CSC): The initial investment cost of various generation technologies includes the one-time expenditures during the early stage of project development, such as construction and equipment procurement costs.。

$$CSC = \sum C_{\mathrm{u}} \varepsilon_{\mathrm{u}} \tag{21}$$

where C_{u} is the additional installed capacity of different types of units, and ε_{u} is the unit cost of additional installed capacity.

System operating costs (COS): The cost invested to ensure the normal operation of the generation system, as shown in Eq. (1).

2.3.2. System flexibility indicators

Wind-solar curtailment rate (WSCR): This indicator represents the ratio of curtailment power to actual wind and solar generation.

Table 1
Operational parameters of FSU and VSU.

Parameter name	Unit type	
	FSU	VSU
Turbine/pump capacity (MW)	300	300
Turbine/pump operating efficiency (%)	85/92	88/92
Pump adjustment range (%)	100	70–100
Minimum stable output power (%)	55	40
Ramping rate (%)	50	100
Start-up and shutdown time (h)	<0.1	<0.1

Table 2
Cost parameters for different types of units.

Unit type	Startup and shutdown costs (CNY/MW)	Ramping costs (CNY/MW)	Construction costs (million CNY/MW)	Fixed O&M costs (CNY/MW)	Variable costs (CNY/MW)
Solar power	0	0	4.6	0.01	133
Wind power	0	0	7.6	0.144	149
Subcritical coal	300	13	4.35	0.07	59
Ultra-supercritical coal-fired	1100	40	3.24	0.07	59
FSU	0	0	6	–	–
VSU	0	0	6.42	–	–

$$WSCR = \left(\sum_i E_i^A \Big/ \sum_i \left(E_i^W + E_i^S \right) \right) \times 100\% \tag{22}$$

where E_i^W and E_i^S are the actual wind and solar generation, respectively.

Wind-solar curtailment power (WSCP): This indicator represents the total curtailment power of the system.

$$WSCP = \sum_i E_i^A \tag{23}$$

Power supply rate (PSR): This indicator represents the ratio between the effective power supply of the system and the power transmission demand. A higher PSR indicates a stronger ability of the system to provide stable power supply, and a value greater than 98 % is considered acceptable.

$$PSR = \left(1 - \sum_i E_i^{LL} \Big/ \sum_i E_i^T \right) \times 100\% \tag{24}$$

where E_i^T is the total transmission demand during the statistical period.

Power supply reliability (PSL): This indicator represents the ratio of the effective power transmission hours (without load shedding) to the total power transmission hours during the statistical period. A value greater than 95 % is considered acceptable.

$$PSL = \left(1 - \sum LH \Big/ \sum TH \right) \times 100\% \tag{25}$$

where $\sum LH$ is the number of load shedding hours, and $\sum TH$ is the total number of hours, which corresponds to 8760 h in this case.

Wind-solar availability rate (WSAR): This indicator represents the ratio of actual wind and solar generation to the available wind and solar generation. A higher WSAR indicates that the power system can absorb wind and solar resources to a greater extent.

$$WSAR = \left(\sum_i \left(E_i^W + E_i^S \right) \Big/ \sum_i \left(E_i^{AW} + E_i^{AS} \right) \right) \times 100\% \tag{26}$$

where E_i^{AW} and E_i^{AS} are the available wind and solar generation, respectively.

Carbon Emissions (CES): This indicator is related to the type and output power of coal power units. Different units with varying installed capacities have different emission coefficients. The carbon emissions for each unit are calculated according to Eq. (20).

2.3.3. Regulation unit operational characteristic indicators

Total number of adjustments of regulation units (TNARU): This indicator includes the number of start-up and shutdown events and load increase and decrease of PS and coal power units. The number of load adjustments reflects the flexibility contribution of the units. When the required adjustment range exceeds the regulation limits of a unit, the adjustment must be achieved through start-up and shutdown operations.

$$TNARU = N_{ST} + N_{LD} + N_{ST}^{coal} + N_{LD}^{coal} \tag{27}$$

Number of start-up and shutdown of PS units (N_{ST}): When the product of PS unit power at two adjacent time periods is negative, it is counted as one start-up and shutdown event of the unit.

$$\begin{cases} \Delta\varepsilon = P_{u,i+1} \times P_{u,i} \\ N_{ST} = \sum_{u,i} Num(\Delta\varepsilon < 0) \end{cases} \tag{28}$$

Number of load increase and decrease of PS units (N_{LD}): By performing a first-order difference on the output between adjacent hours, the output variation $\Delta P_{u,i}^G$ of each unit is obtained. When $\Delta P_{u,i}^G$ is not equal to zero, it indicates that the unit has completed one load adjustment. Since FSU can only operate with a fixed input during pumping, the number of load increase and decrease actions is counted only during the generation state.

$$\begin{cases} \Delta P_{u,i}^G = P_{u,i+1}^G - P_{u,i}^G \\ N_{LD} = \sum_{u,i} Num\left(\Delta P_{u,i}^G \neq 0 \right) \end{cases} \tag{29}$$

Number of start-up and shutdown of coal power units (N_{ST}^{coal}): If the output of a coal power unit is zero at the previous time period and greater than zero at the next time period, it is counted as one start-up. Conversely, if the output decreases from greater than zero to zero, it is counted as one shutdown.

$$N_{ST}^{coal} = \sum_{u,i} Num\left(\left(P_{u,i+1}^{coal} = 0, P_{u,i}^{coal} > 0 \right) + \left(P_{u,i+1}^{coal} > 0, P_{u,i}^{coal} = 0 \right) \right) \tag{30}$$

Number of load increase and decrease of coal power units (N_{LD}^{coal}): The calculation method is similar to that of the load increase and decrease of PS units and is determined by Eq. (29).

Cumulative power regulation (CPR): To quantify the power fluctuation regulation capability of PS, the cumulative power regulation indicator is introduced. It is calculated by summing the absolute values of the power differences between adjacent hours. The specific calculation formula is:

$$CPR = \sum_{u,i} \left| P_{u,i+1} - P_{u,i} \right| \tag{31}$$

Cumulative number of power regulation (CNPR): If the power output of a PS unit differs between two consecutive hours, it is counted as one instance of cumulative power regulation.

$$\begin{cases} \Delta P_{u,i} = P_{u,i+1} - P_{u,i} \\ CNPR = \sum_{u,i} Num(\Delta P_{u,i} \neq 0) \end{cases} \tag{32}$$

2.4. Comprehensive evaluation method

This study proposes a comprehensive evaluation method based on the entropy weight-TOPSIS approach to quantitatively assess the flexibility advantages and overall benefits of FSU and VSU within clean energy bases. The method integrates the objective weighting capability of the entropy weight method with the multi-attribute decision-making strength of the TOPSIS method [37], effectively addressing complex multi-indicator evaluation problems. The evaluation process is illustrated in Fig. 2.

Step 1: First, the data for each indicator are subjected to positive transformation and normalization.

The positive transformation formula can be expressed as follows:

$$\begin{cases} x_{ij} = \dfrac{x_{ij0} - \min(x_j)}{\max(x_j) - \min(x_j)} \\ x_{ij} = \dfrac{\max(x_j) - x_{ij0}}{\max(x_j) - \min(x_j)} \end{cases} \tag{33}$$

where x_{ij} is the j-th indicator of the i data group, and $\max(x_j)$ and $\min(x_j)$ are the maximum and minimum values of the j indicator, respectively.

The Z-score method is used to normalize each indicator, and the specific formula is as follows:

$$x_{ij}^{'} = \frac{x_{ij} - mean(x_j)}{std(x_j)} \tag{34}$$

where $mean(x_j)$ is the mean value of x_j, and $std(x_j)$ is the standard deviation of x_j.

Step 2: Calculate the weight of each indicator based on its internal information entropy.

For indicator r_j, the information entropy is calculated as:

$$E_j = -\frac{1}{\ln m}\sum\nolimits_{i=1}^{m} p_{ij}\ln p_{ij} \tag{35}$$

where p_{ij} is an element of the probability matrix, and its expression is:

$$p_{ij} = \frac{r_{ij}}{\sum_{i=1}^{m} r_{ij}} \tag{36}$$

The information utility values are normalized, and the weights of each indicator are obtained as follows:

$$W_j = \frac{(1 - E_j)}{\sum_{j=1}^{n}(1 - E_j)} \tag{37}$$

Step 3: The weighted data are denoted as r_{ij}, forming the data matrix $R = (r_{ij})_{m\times n}$, and the maximum and minimum values for each indicator are calculated as follows:

$$r_{ij} = w_j x_{ij}^{'} \tag{38}$$

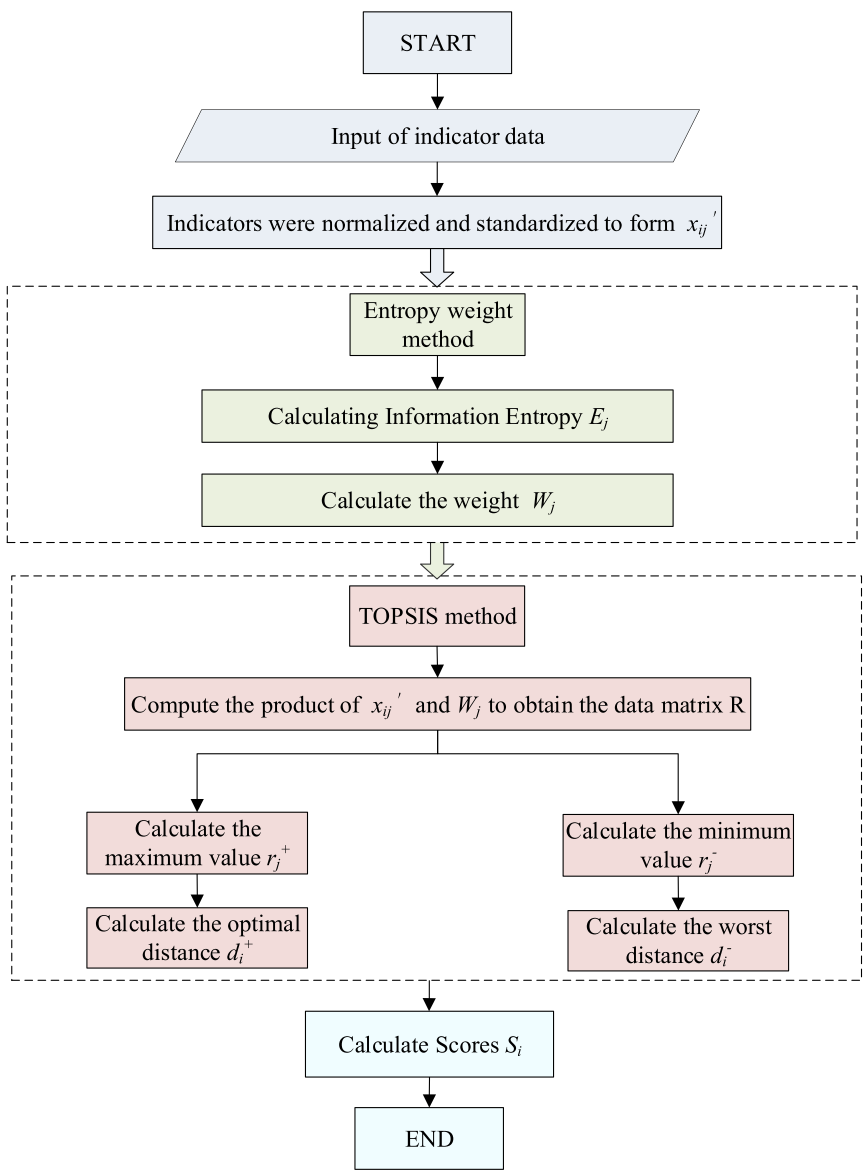


Fig. 2. Entropy weight-TOPSIS method evaluation process.

$$r_j^+ = \max(r_{1j}, r_{2j} \ldots, r_{nj}) \tag{39}$$

$$r_j^- = \min(r_{1j}, r_{2j} \ldots, r_{nj}) \tag{40}$$

Step 4: Calculate the optimal and worst distances for each data group:

$$d_i^+ = \sqrt{\sum_{j=1}^{n} (r_j^+ - r_{ij})^2} \tag{41}$$

$$d_i^- = \sqrt{\sum_{j=1}^{n} (r_j^- - r_{ij})^2} \tag{42}$$

Step 5: Calculate the final score:

$$S_i = \frac{d_i^-}{d_i^+ + d_i^-} \tag{43}$$

2.5. Case study

2.5.1. Area overview

The study area of this paper is located in Qinghai Province, northwestern China—a region abundant in clean energy resources with significant potential for renewable energy development [30,38]. In recent years, the large-scale construction of PS plants has steadily increased the clean energy accommodation rate in northwest China. However, the operation of PS plants in this region faces two major challenges. First, as most clean energy bases are situated in remote areas, they suffer from limited local capacity to accommodate clean energy and a shortage of regulation resources, placing immense pressure on grid integration and operational reliability. Second, PS units serve as complementary sources within clean energy bases and are increasingly required to provide enhanced flexibility to ensure the stability and security of long-distance power transmission.

Effectively leveraging the flexibility provision capability of PS based on regional renewable energy layouts and flexibility demands has become a key issue for the long-term development of PS. To address this, a clean energy base in Qinghai Province is selected as a case study. Based on the planned installed capacities, an energy base unit combination model is developed. Hourly renewable energy output time series are generated using local meteorological data, and a full-year operation simulation over 8760 h is carried out, incorporating the planned power transmission curve. The operational characteristic differences between FSU and VSU within the clean energy base are then compared. The power transmission curve of the clean energy base is shown in Fig. 3.

2.5.2. Scenario classifications

By adjusting the wind–solar installed capacity ratio in the renewable energy mix and the type of PS units, this study defines several typical scenarios. Three representative scenarios are selected as case studies to comprehensively evaluate the flexibility performance of FSU and VSU under different wind–solar ratio conditions. The installed capacities for each scenario are listed in Table 3. In all scenarios, the PS plants is

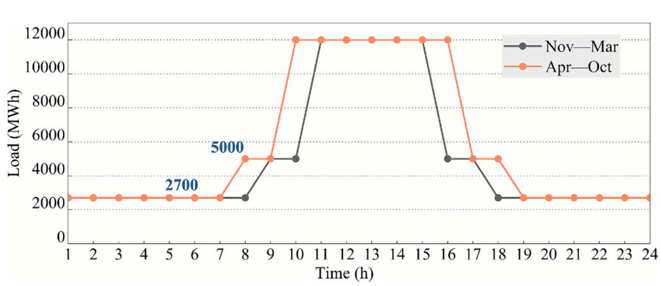


Fig. 3. Power transmission curve of the clean energy base.

modeled with a rated generating capacity of 2400 MW and a usable storage capacity of approximately 24 GWh, corresponding to about 10 h of continuous full-load generation at rated power.

2.5.3. Wind and solar available factor

The actual power generation capability of the photovoltaic station is characterized by the solar availability factor. Based on data from the Global Solar Atlas (https://globalsolaratlas.info/map), hourly solar availability factors can be obtained by entering the specific coordinates of the photovoltaic base (95.16°, 36.39°), as shown in Fig. 4(a). Wind availability factor data are sourced from the Global Renewable Energy Atlas (https://www.renewables.ninja/). By inputting the coordinates of the wind power base (95.72°, 36.59°), the corresponding local wind availability factor time series can be retrieved, as shown in Fig. 4(b).

3. Results

3.1. System flexibility evaluation results

According to the transmission plan, this study selects five typical days in January, representing the low-step power transmission period, and five typical days in August, representing the high-step power transmission period, to comparatively analyze output power characteristics under different scenarios. The complementary effects among various energy sources under different wind-solar ratios, as well as the balance between system output and the power transmission curve, are examined. The output power profiles of each energy source under different scenarios are shown in Fig. 5.

In Fig. 5, panels (a) and (c) correspond to scenarios configured with FSU, while panels (b) and (d) represent scenarios configured with VSU. It can be observed that in the power transmission system, coal power primarily serves as the nighttime baseload supplier and partially undertakes regulation tasks. PS plants store energy during periods of high wind and solar output and coordinate with wind and coal power to meet transmission demand during nighttime hours when photovoltaic output is unavailable. When the output from renewable and coal power is insufficient, PS switches to generation mode to perform regulation tasks. According to the January transmission curves, the short duration of high-step periods results in significantly lower overall transmission demand compared to August, leading to noticeable curtailment of wind and solar power. In August, curtailment is substantially reduced; however, there are instances when peak load demand cannot be fully met, as indicated by the pink shaded areas in the figure.

In the S2 and S3 scenarios, which feature a higher share of wind power, PS units frequently alternate between pumping and generation modes to provide regulation services. Load shedding becomes more severe as the wind power proportion increases. This phenomenon is particularly evident in January. With rising wind power penetration, coal power generation decreases significantly, and output fluctuations become more pronounced due to the reduction in coal power output. In August, owing to higher transmission demand, coal power primarily undertakes the baseload task to meet system requirements, and its output remains relatively stable. Regarding the differences between FSU and VSU performance, on January 30 under the S2 scenario and August 30 under the S3 scenario, VSU significantly mitigates wind and solar power curtailment. However, with respect to load shortages, which constitute a relatively small portion of total generation, it is difficult to draw definitive conclusions based solely on Fig. 5. Further statistical analysis is required.

3.1.1. Load shedding

The intermittent characteristics of wind and solar output power, combined with insufficient upward regulation flexibility of the units, introduce a risk of power supply shortages, resulting in load shedding that reduces the system's stability and reliability. Fig. 6 presents the median, mean and annual total load-shedding values. Each hollow circle

Table 3
Configuration of multi-energy complementary scenarios.

Scenario	Ultra-supercritical coal-fired (MW)	Subcritical coal (MW)	Pump storage (FSU/ VSU) (MW)	Wind power (MW)	Solar power (MW)	Wind/solar capacity ratio	Renewable-energy share
S1	1000 × 2	300 × 2	300 × 8	1590	15,910	10 %	78 %
S2	1000 × 2	300 × 2	300 × 8	2917	14,583	20 %	78 %
S3	1000 × 2	300 × 2	300 × 8	4038	13,462	30 %	78 %

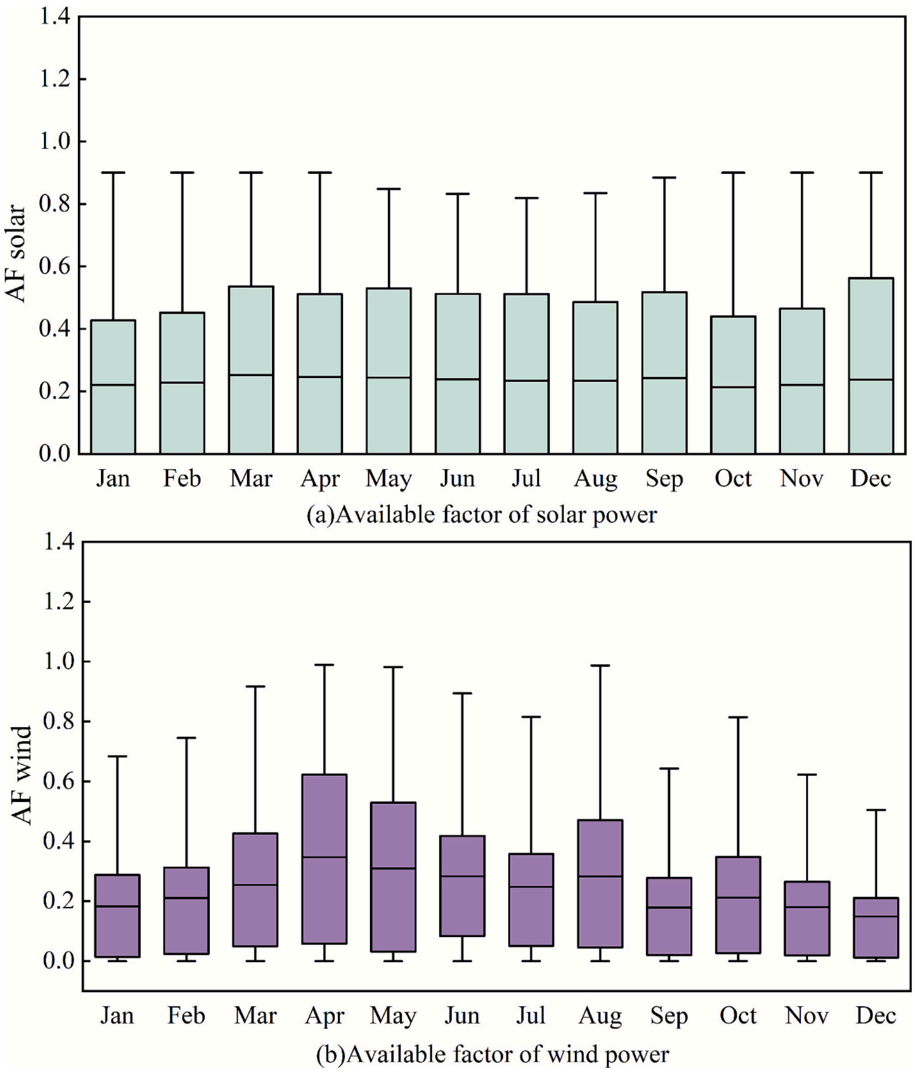


Fig. 4. Distribution map of annual available factor of wind power and solar power.

represents the corresponding monthly load-shedding value.

As shown in Fig. 6, regardless of whether the system is configured with FSU or VSU, the annual, median, and mean values of load shedding all exhibit an increasing trend with the rising proportion of wind power. Specifically, for FSU, the annual load shedding in the S3 scenario increases by 168.35 GWh compared to S1, while for VSU, the increase is 164.19 GWh. In terms of total load shedding, the ranking across the three scenarios follows the order S1 < S2 < S3. This trend is attributed to the reduced share of photovoltaic power, which results in greater load deficits during periods of low solar irradiance. During such times, the power output from wind units may be lower than that of photovoltaic units with the same installed capacity.

Comparing the performance of FSU and VSU, in the S1 scenario, the annual load shedding of VSU is 20.58 GWh lower than that of FSU, and in the S3 scenario, the reduction reaches 24.74 GWh. This indicates that a higher share of wind power introduces greater random fluctuations. VSU, owing to its superior ramping capability, can alleviate load shedding caused by increased wind penetration. However, due to capacity limitations, its effectiveness remains inferior to the improvement achieved through increased photovoltaic installed capacity. This highlights that the highly stochastic nature of wind power output significantly affects the reliability of system transmission and intensifies load shedding issues. Therefore, when the flexibility resources within a multi-energy complementary system are insufficient to meet operational demands, it becomes necessary to further optimize the power transmission curve or deploy additional energy storage facilities as appropriate in order to ensure the effective provision of system flexibility.

3.1.2. *Wind-solar curtailment rate*

Due to the limited local capacity to accommodate energy within the

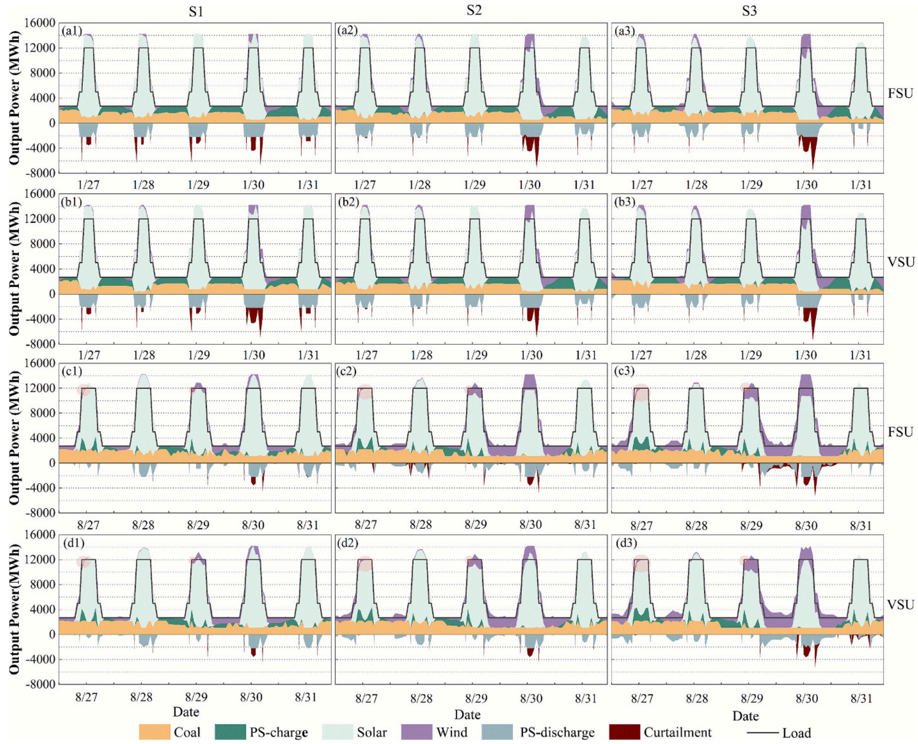
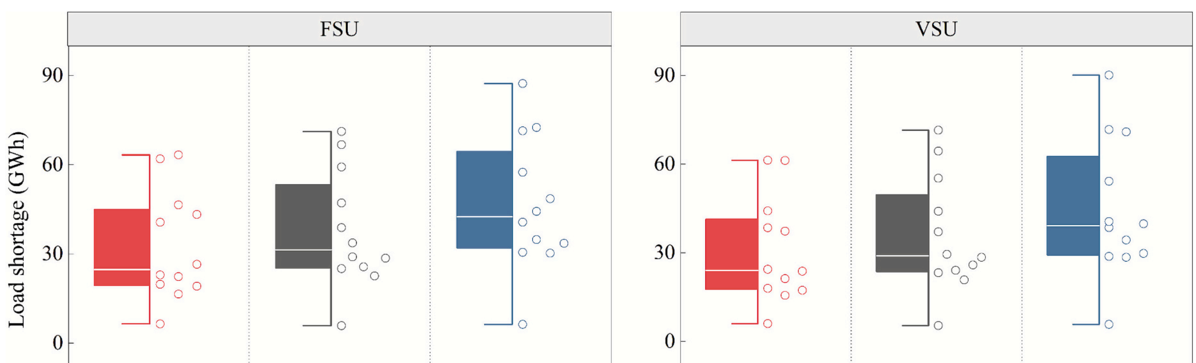


Fig. 5. Hourly power generation of each energy source on typical days in different configuration scenarios.



	FSU S1	FSU S2	FSU S3	VSU S1	VSU S2	VSU S3
Median	24.74	31.36	42.53	24.09	29.01	39.19
Mean	32.47	37.83	46.50	30.75	35.82	44.44
Total	389.63	453.92	557.98	369.05	429.85	533.24

Fig. 6. Annual load shortage statistics in different scenarios.

clean energy base and the transmission constraints imposed by the power transmission curves, wind-solar curtailment remains relatively severe. Fig. 7 quantitatively illustrates the annual wind-solar curtailment rates and curtailment power of both FSU and VSU under different scenarios.

As shown in Fig. 7, with the increase in wind power proportion, both the annual wind-solar curtailment rates and the annual curtailment power exhibit a decreasing trend across scenarios, with a more pronounced decline from S1 to S2. Specifically, for FSU, the annual curtailment power decreases by 675.21 GWh and the curtailment rate drops by 2.05 %; for VSU, the annual curtailment power decreases by 420.37 GWh and the curtailment rate by 1.29 %. The curtailment issue in the power transmission system is partially alleviated with higher wind power penetration, primarily because the reduction in photovoltaic share leads to fewer curtailment events during daytime hours—periods that, as seen in Fig. 5, account for most curtailment occurrences. In terms of performance comparison, VSU demonstrates superior capability in reducing wind-solar curtailment. This is particularly evident in the S1 scenario, where VSU achieves 753.41 GWh less curtailment power than FSU and reduces the curtailment rate by 2.2 %.

3.1.3. *Power supply rate and power supply reliability*

To comprehensively evaluate the transmission performance and power supply quality of the energy base, the system's power supply rate and reliability under different scenarios are summarized in Fig. 8.

As shown in Fig. 8, whether configured with FSU or VSU, both the power supply rate and power supply reliability of the system decline as the proportion of wind power increases. However, the decrease is significantly smaller for VSU compared to FSU. This indicates that under higher wind power penetration, configuring VSU can more effectively

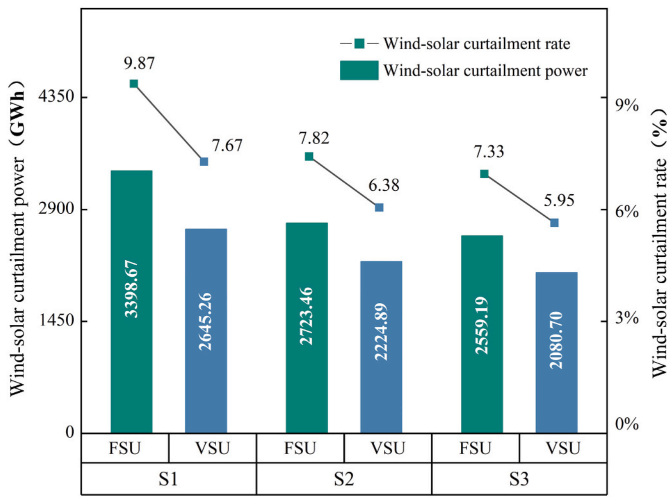


Fig. 7. Annual wind-solar curtailment rate and curtailment power under different scenarios.

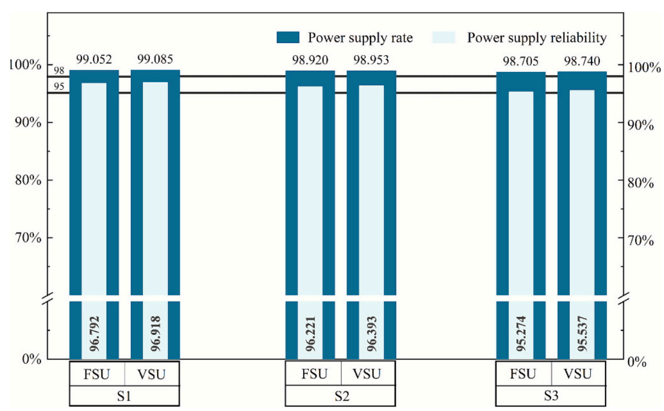


Fig. 8. Power supply rate and reliability in different scenarios.

meet the load demand of the transmission system and significantly enhance the overall power supply security.

In all three scenarios, the power supply rate exceeds 98 %, with the best performance observed in the S1 scenario with VSU, reaching 99.085 %, and the worst in the S3 scenario with FSU, at 98.705 %. Similarly, power supply reliability in all scenarios surpasses 95 %, with the highest value of 96.918 % in the S1 scenario with VSU and the lowest of 95.274 % in the S3 scenario with FSU. It is important to note that the minimum requirements for the energy base are a power supply rate above 98 % and power supply reliability above 95 %, and all scenarios analyzed meet these criteria.

3.1.4. Wind-solar availability rate

The higher the wind-solar availability rate, the greater the utilization of wind and solar resources, leading to a reduced burden on grid dispatching. The wind-solar availability rate of the power system under different scenarios is shown in Fig. 9.

As shown in Fig. 9, different wind-solar capacity ratios have a significant impact on the wind-solar availability rate of the system. With the increasing proportion of wind power, the system's wind-solar availability rate shows an overall upward trend. From the S1 to S3 scenarios, the availability rate increases by 1.25 % for FSU and by 1.27 % for VSU. This improvement is primarily attributed to the fact that, compared to photovoltaics, wind power output exhibits greater continuity and can effectively compensate for the absence of photovoltaic generation during nighttime hours, thereby significantly enhancing the

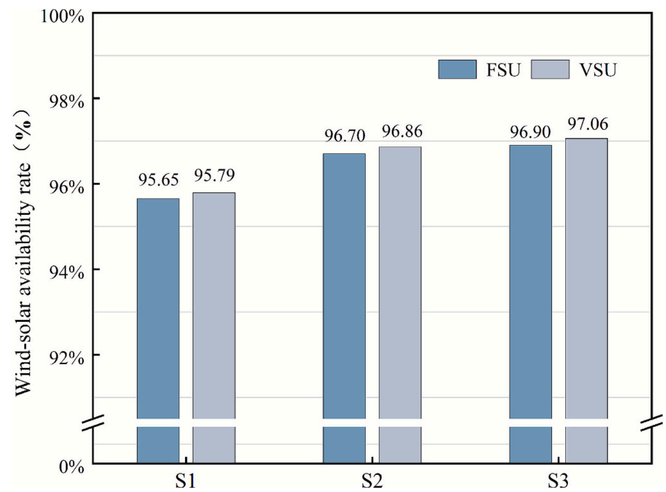


Fig. 9. Wind-solar availability rates in different scenarios.

system's ability to utilize wind and solar resources throughout the day.

Further analysis reveals that in all three scenarios, systems configured with VSU exhibit a higher wind-solar availability rate. In the S1 scenario, the availability rate of VSU is 0.14 % higher than that of FSU, while in both the S2 and S3 scenarios, it increases by 0.16 %. This improvement is primarily due to the more flexible and precise load regulation capability of VSU, which enables it to effectively smooth fluctuations in wind and solar output, thereby further enhancing the overall utilization of wind and solar resources within the power system.

3.1.5. Total power generation

Different wind-solar ratios influence the output power of each unit, and over time, significant differences in total power generation among the various units emerge. Fig. 10 presents the annual total output power (GWh) of four energy sources: coal power, PS, solar power, and wind power across the three scenarios.

As shown in Fig. 10, solar power and coal power contribute the most to the total output of the energy base. With increasing wind power penetration, the share of wind power output rises, which not only reduces photovoltaic output but also gradually lowers coal power generation. VSU effectively increases the amount of wind power integrated into the grid, further decreasing coal power output. Under the same scenario, the configuration of VSU results in a slight increase in wind power output in the S1 and S3 scenarios, while a slight decrease is observed in S2. Photovoltaic output slightly decreases in S1 and S3 but increases significantly in S2. Overall, the system's total clean energy penetration is effectively enhanced.

With the continuous increase in wind power proportion, the generation output of PS units exhibits a gradual declining trend, with a maximum reduction of 204.1 GWh for FSU and 214 GWh for VSU. Across all three scenarios, VSU consistently produces more output than FSU under the same conditions, with the largest difference occurring in the S3 scenario, where VSU exceeds FSU by 201.3 GWh.

3.1.6. Carbon emissions

Coal power units, which are responsible for most of the baseload tasks in the system, have a significant impact on the low-carbon objectives of the energy base. Excessive carbon emissions increase both economic and environmental costs. The annual total carbon emissions under different scenarios are summarized in Table 4.

By analyzing Table 4 and Fig. 10, it can be observed that carbon emissions are closely linked to the output of coal power units, and the system's carbon emissions exhibit a decreasing trend as the proportion of wind power increases. This is mainly because coal power primarily provides baseload supply during nighttime, while the continuous output

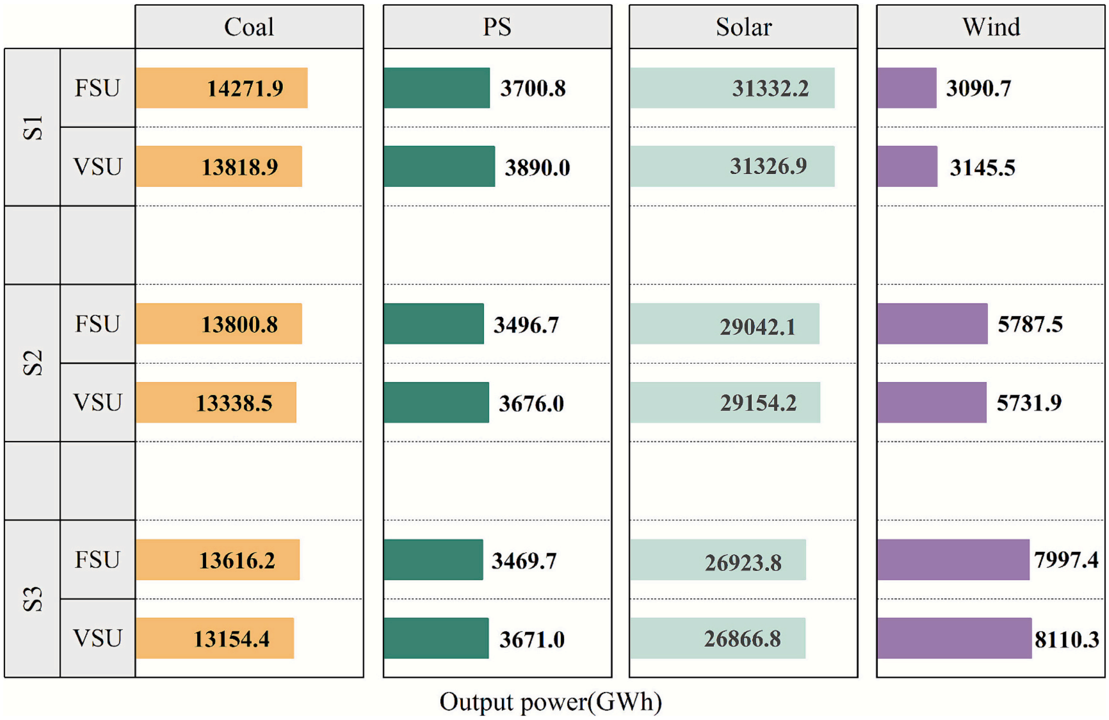


Fig. 10. Statistics of unit power generation in different scenarios.

Table 4
Carbon emissions in different scenarios.

Scenario	Unit type	Carbon emission (tonne)
S1	FSU	14,985.52
	VSU	14,509.81
S2	FSU	14,490.81
	VSU	14,005.44
S3	FSU	14,297.04
	VSU	13,812.14

of wind power during these hours effectively replaces part of the coal baseload, thereby significantly reducing the system's overall carbon emissions.

Further comparison of carbon emissions between different PS unit types reveals that in the S1 scenario, the system configured with VSU reduces carbon emissions by 475.71 t compared to FSU, representing a 3.17 % decrease. In the S2 and S3 scenarios, emissions are reduced by 485.4 t and 484.9 t, corresponding to reductions of 3.35 % and 3.4 %, respectively. This demonstrates that VSU, owing to its faster response speed and broader regulation range, can more effectively accommodate wind power fluctuations, reduce the regulation burden on coal power units, and thereby significantly lower the overall carbon emissions of the power system.

3.2. System economic evaluation results

3.2.1. Construction cost and system operating cost

To evaluate the economic feasibility and benefits of FSU and VSU, this study adopts construction cost and system operating cost as evaluation indicators. The differences between the two unit types are analyzed in terms of both construction and operational expenses, aiming to provide a scientific basis for decision-making. The results are presented in Fig. 11.

The construction cost in each scenario is determined by the installed capacity and the unit construction cost. As the proportion of wind power increases, construction costs show an upward trend across the scenarios.

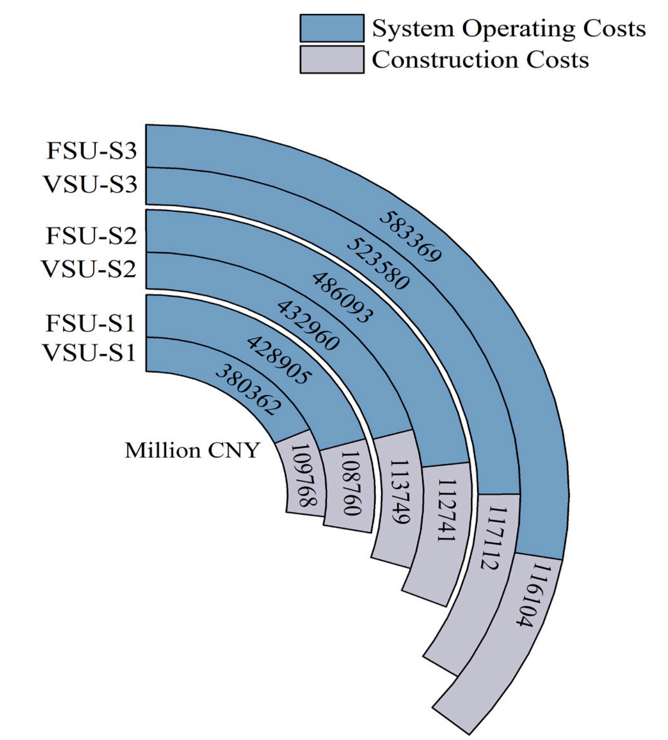


Fig. 11. Construction and operating costs of energy systems in different scenarios.

The energy system configured with VSU generally incurs higher construction costs than that with FSU, with a difference of approximately 1 billion CNY. The system operating cost is influenced by both the wind-solar capacity ratio and the type of pumped storage technology. With the increasing share of wind power, the operating cost also rises, reaching a difference of 143 billion CNY between the VSU S3 and VSU S1 scenarios.

It is worth noting that although the construction cost of VSU is higher than that of FSU, the system operating cost is significantly lower for VSU, with the largest difference observed in the S3 scenario, reaching approximately 60 billion CNY. This is primarily because the power system configured with VSU demonstrates significantly enhanced flexibility, effectively reducing load shedding and wind-solar curtailment, thereby substantially lowering the system's operating cost.

3.3. Evaluation results of regulation unit operational characteristics

To compare the regulation range and the main regulation intervals of the units during the load adjustment process, Fig. 12 presents the probability density distribution of output power fluctuations for FSU and VSU under different scenarios.

As shown in Fig. 12, the wind-solar capacity ratio has a relatively minor impact on unit output power fluctuations, whereas the type of pumped storage unit leads to more pronounced differences. In scenarios configured with FSU, output power fluctuations are concentrated within the range of [±155 MW, ±175 MW], while in VSU-configured scenarios, the fluctuations are concentrated within [±110 MW, ±135 MW]. A comparison of FSU and VSU in the S1 scenario shows that the fluctuation density of FSU is generally higher, indicating more concentrated variations within the main fluctuation interval. VSU, due to its ability to adjust output power more precisely and rapidly, effectively reduces grid load fluctuations with a smaller fluctuation range, resulting in overall higher regulation efficiency.

3.3.1. Cumulative power regulation and cumulative number of power regulation of PS units

To quantitatively compare the flexibility contributions of the two types of units, Fig. 13 presents the cumulative power regulation and the cumulative number of power regulation of PS units.

As shown in Fig. 13, with the increasing proportion of wind power, both the cumulative power regulation and the cumulative number of regulation actions for FSU and VSU exhibit an upward trend. This indicates that PS units are required to undertake more regulation tasks as wind power penetration rises. From the S1 to S3 scenario, the cumulative power regulation of FSU increases by 910.34 GWh, representing an 8.25 % growth, while the cumulative number of regulation actions increases by 2315, corresponding to a 7.94 % increase. For VSU, the cumulative power regulation increases by 627.73 GWh with a growth rate of 7.63 %, and the cumulative number of regulation actions increases by 1388 with a growth rate of 4.13 %.

It can be observed that although the growth rates of cumulative power regulation and the cumulative number of regulation actions for VSU are lower than those of FSU, the overall cumulative number of regulation actions for VSU remains higher. Compared to FSU, VSU features faster ramping speed and a broader operating range. As a result,

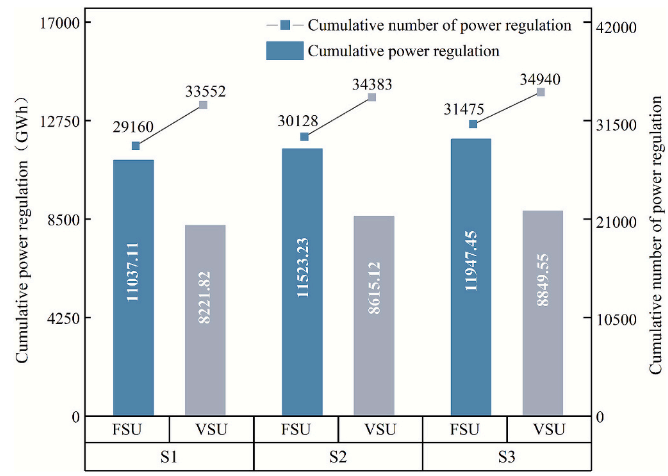


Fig. 13. Cumulative power regulation and cumulative number of power regulation of PS units in different scenarios.

VSU performs regulation actions more frequently, but with smaller amplitudes. This leads to a relatively lower cumulative power regulation, while allowing VSU to more effectively smooth system load fluctuations.

3.3.2. Total number of adjustments of regulation units

Frequent participation of coal power and PS units in load regulation increases equipment wear and operating costs. A higher number of adjustments reduces the operational stability and economic efficiency of the power supply. Table 5 summarizes the total number of adjustments in the entire power system.

As shown in Table 5, the total number of system adjustments increases for both FSU and VSU as the proportion of wind power rises. This indicates that higher wind power penetration imposes greater flexibility requirements on the power system, requiring regulation units to carry out more load adjustments and more frequent start-up and shutdown operations to maintain system stability.

In scenarios configured with VSU, the total number of adjustments shows a noticeable reduction. Across all three wind-solar ratio scenarios, the total number of adjustments with VSU is consistently lower than that with FSU. Specifically, in the S1 scenario, the total number of adjustments decreases by 4712, representing a 12.12 % reduction; in the S2 scenario, it decreases by 3658, corresponding to an 11.97 % reduction; and in the S3 scenario, it decreases by 5781, with a reduction rate of 13.9 %. The results indicate that VSU, due to its superior regulation flexibility, reduces the need for coordinated adjustments among various types of regulating units. This effectively alleviates the regulation burden on coal power units, smooths their output power, and decreases

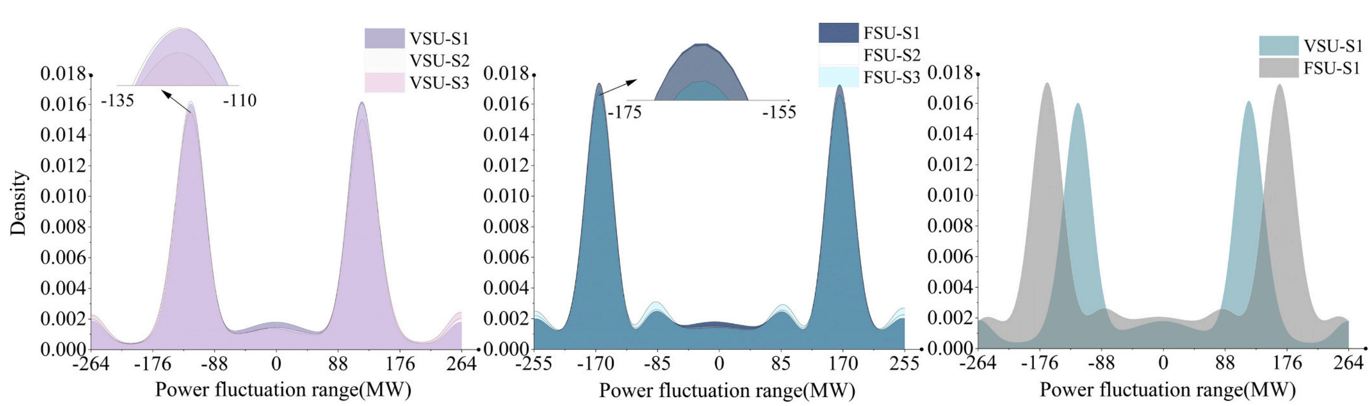


Fig. 12. Density curves of PS unit output fluctuations under different scenarios.

Table 5
Total number of adjustments of regulation units in different scenarios.

Regulatory indicators	S1		S2		S3	
	FSU	VSU	FSU	VSU	FSU	VSU
Number of load increase and decrease of PS	5442	6447	5421	6362	6059	6783
Number of start-up and shutdown of PS	23,718	19,831	24,707	20,954	25,416	21,160
Number of load increase and decrease of coal power	9254	7459	9679	7721	9530	7291
Number of start-up and shutdown of coal power	453	418	546	483	585	575
Total number of adjustments of regulation units	38,867	34,155	40,353	35,520	41,590	35,809

the frequency of start-up and shutdown operations. Consequently, it not only preserves the overall regulation capability of the power system but also lowers the operating costs of the entire energy system.

It is worth noting that although the number of load adjustments for VSU is higher than that for FSU, the number of start-up and shutdown operations is significantly lower. This is because VSU, with its wider regulation range, can accommodate load variations through continuous power adjustments. In contrast, FSU typically requires a combination of start-up/shutdown operations and inter-unit load adjustments to achieve the same level of regulation, often resulting in longer adjustment times.

3.4. Comprehensive evaluation

To quantify the flexibility differences between FSU and VSU within the energy system, this section employs the entropy weight-TOPSIS method to conduct a comprehensive benefit evaluation across three dimensions: system economic performance, system flexibility, and operational characteristics of regulation units. The indicator weights and evaluation scores for FSU and VSU under different wind power penetration scenarios are subsequently determined.

As shown in Fig. 14, system flexibility indicators carry the highest weight (51.92 %), followed by regulation unit operational characteristic indicators (26 %) and system economic indicators (22.08 %). The highest weights among all individual indicators are held by CPR (14.25 %), CSC (12.77 %) and PSL (12.38 %), corresponding respectively to the dimensions of regulation unit operational characteristics, system economics and system flexibility. This distribution is a direct outcome of the entropy-weighting procedure, in which indicators with greater dispersion across scenarios (lower entropy and stronger discriminative power) are assigned larger weights, while more homogeneous indicators receive smaller weights. Consequently, the comprehensive TOPSIS scores are most sensitive to variations in flexibility-related indices, indicating that the proposed entropy weight–TOPSIS evaluation framework places greater emphasis on system flexibility performance.

Based on the indicator weights, Fig. 15 presents the comprehensive evaluation scores for all scenarios, assessed on a 100-point scale.

Fig. 15 presents annular charts of the comprehensive benefit scores for each scenario. The results for different PS technologies indicate that the average score for VSU-configured scenarios is 88.95, significantly higher than the 47.99 recorded for FSU-configured scenarios. In all three scenarios, VSU consistently outperforms FSU. Notably, in the S2 scenario, the advantages of VSU are most evident, with the system flexibility score increasing by 23.02 points, and the scores for regulation unit operational characteristics and system economics increasing by 11.51 and 9.79 points, respectively.

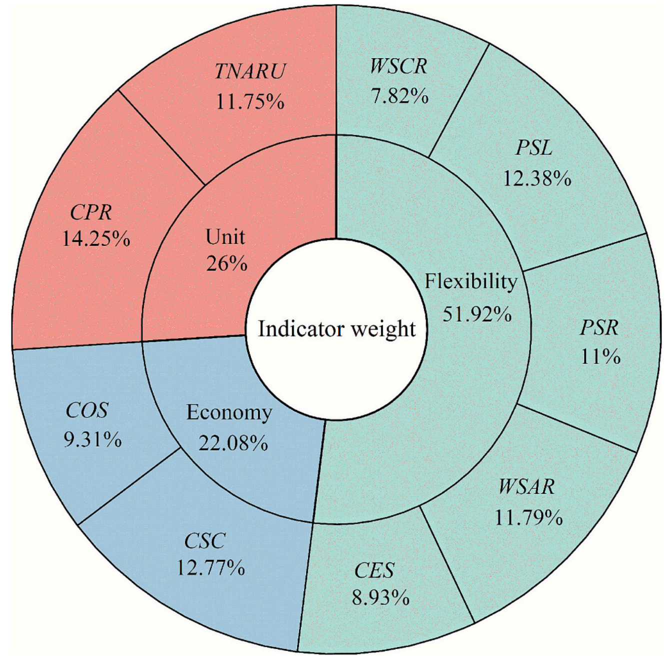


Fig. 14. The weight coefficients of each evaluation index.

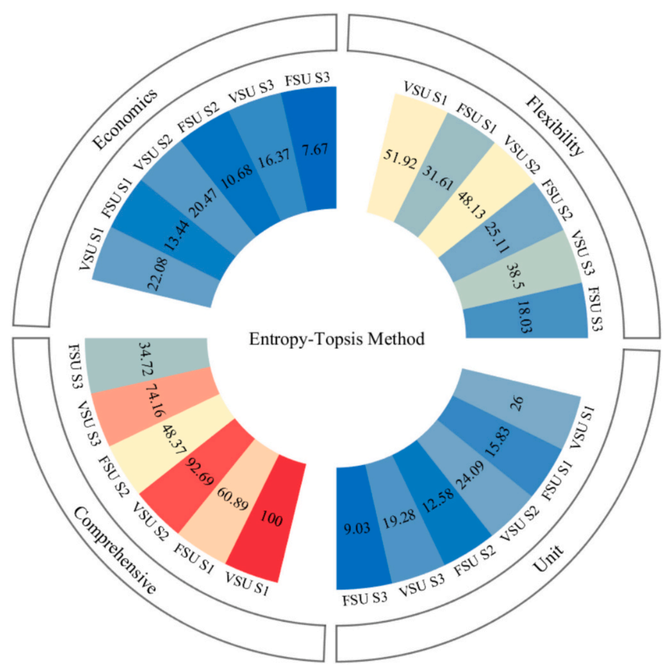


Fig. 15. The evaluation result of comprehensive benefits.

With the increase in wind power penetration, the scores of both FSU and VSU exhibit a downward trend. From the S1 to S3 scenarios, the score of VSU decreases by 25.84 points, while that of FSU declines more significantly by 26.17 points. This suggests that the high variability of wind power output imposes considerable challenges on the overall energy system. However, due to its faster regulation speed and broader adjustment range, VSU can more effectively enhance system flexibility and reduce operating costs, thereby rendering its comprehensive benefits more prominent.

4. Conclusion

With the increasing share of renewable energy capacity in clean energy bases, the demand for flexibility from PS units in power systems has become increasingly significant. This study investigates the differences in operational characteristics between FSU and VSU within a multi-energy complementary system and quantitatively evaluates the flexibility advantages of VSU in clean energy bases. Firstly, by comprehensively considering the regulatory characteristic differences between FSU and VSU, a multi-energy complementary unit combination model incorporating various types of generation units is developed. Second, from the perspectives of system flexibility, economic performance, and regulation unit operational characteristics, a set of evaluation indicators is proposed to assess the comprehensive benefits of the energy system. Finally, using a DC power transmission-based clean energy base in Qinghai Province as a case study, the entropy weight-TOPSIS method is employed to evaluate the operational performance differences between FSU and VSU under flexible operating modes. The main conclusions are as follows:

(1) The increase in the proportion of wind power reduces coal power output and promotes the low-carbon transition of the energy system. From scenario S1 to scenario S3, the CES and WSCR indicators gradually decline, while the WSAR indicator steadily increases. However, the high variability of wind power significantly affects transmission reliability, intensifying the load shedding problem within the transmission system. Consequently, PSR and PSL both decrease, and the system's flexibility demand rises. In this case study, for every 10 % increase in wind power share, PSL declines by approximately 1 %. Additionally, as the wind power share increases, the CPF and TNARU indicators increase by around 5 %.

(2) Based on the differences in response characteristics between FSU and VSU, this study conducts a detailed comparison of their flexibility performance in a multi-energy complementary system by analyzing parameters such as regulation range, efficiency, and ramping capability. Compared to FSU, the system configured with VSU demonstrates slight improvements in PSR, PSL, and WSAR across different scenarios, while WSCR and CES decrease by 2 % and 3 %, respectively. Moreover, under the same scenario, the number of start-up and shutdown operations for VSU is lower than that for FSU. In this case study, the number of start-up and shutdown events for VSU is reduced by more than 15 %, despite a higher cumulative number of power regulation actions than FSU. This is because VSU allows for adjustable input power during pumping mode and features a wider operating range in turbine mode. In situations where FSU requires start-up and shutdown to perform regulation, VSU can instead respond through continuous load adjustments.

(3) Configuring VSU increases grid-connected wind power and further reduces coal power output. The total penetration rate of clean energy in the system has been increased. Coal power remains a key source of regulation in this case study. VSU assumes a larger share of regulation tasks, thereby reducing the need for coordinated adjustments among various regulating units. This effectively alleviates the burden on coal power units, smooths their output profiles, and decreases the frequency of start-up and shutdown operations. While maintaining the system's overall regulation capability, the VSU configuration also lowers the operating costs of the energy system, yielding notable improvements in operational stability and carbon emission reduction.

(4) The comprehensive evaluation results demonstrate that the overall benefits of VSU are significantly greater than those of FSU. Across the three scenarios, the average comprehensive score of VSU reaches 88.95, which is 40.96 points higher than FSU's score of 47.99. Specifically, VSU improves the average score of system flexibility indicators by 21.26 points, while increasing the scores for regulation unit operational characteristics and system economic indicators by 10.64 and 9.04 points, respectively. As wind power penetration rises, the demand for system flexibility grows substantially. VSU, with its faster response and greater operational flexibility, plays a critical role in enhancing system adaptability and improving the overall comprehensive benefits of the energy system.

5. Discussions

This study systematically compares the operational characteristics of FSU and VSU within a multi-energy complementary clean energy base from both technical and economic perspectives. However, certain limitations remain, as the analysis only considers standalone configurations of FSU and VSU, without examining hybrid layouts that combine both unit types. Future research will focus on evaluating the adaptability of different FSU and VSU configuration ratios in PS plants to clean energy bases. Additionally, by incorporating more comprehensive flexibility and economic indicators, future studies will aim to explore optimal hybrid configuration strategies that balance cost-effectiveness and regulation performance, providing more robust theoretical support for practical engineering applications.

CRediT authorship contribution statement

Yunpeng Zhang: Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Gongcheng Liu:** Writing – review & editing, Resources. **Zhang Liu:** Writing – review & editing, Resources. **Meng Zhang:** Writing – review & editing, Resources. **Jianling Li:** Visualization, Resources. **Ziwen Zhao:** Software, Investigation. **Ali Najah Ahmed:** Writing – review & editing, Methodology. **Diyi Chen:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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